Real-World Validation

KudiGuard — NAIC 2026, Problem Statement 2 (Voice-First Access)

Document 3 of the submission: benchmark results on frozen test sets, speech accuracy, and the live pilot of 12 Oct 2026 with what it found and what we fixed the same day. Every number names its source file; the full claims register with commands to reproduce is `docs/CLAIMS.md`.

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1. Summary

Evidence	Result	Source
Frozen end-to-end benchmark **T v1** (366 cases, 5 languages, run once)	pass **85%**; scams flagged **178/196 (91%)**; genuine messages wrongly flagged **35/96 (36%)**; questions answered 72/74; p50 3.5 s, p95 8.3 s; 0 errors	`docs/CLAIMS.md` C10, `data/eval/results/e2e-live.jsonl`
Advice fixes on dev set dev2 (130 cases)	judged good 10 -> 26; advice points 32% -> 46%; scams flagged 48/50 -> 48/50	`docs/CLAIMS.md` C14
NCAIR1 speech, word error rate	Hausa 28%, Yoruba 33%, Igbo 46% (read speech)	`docs/CLAIMS.md` C3
Live pilot, 12 Oct 2026	25 real interactions (20 voice notes) from 5 people (4 external testers + the team lead), Hausa and Yoruba, on the web voice page; team judgement: 11 of 25 handled right (44%), 9 of 16 scam descriptions flagged; 12 issues found, 11 fixed and 1 experiment rejected the same day	`var/kudiguard.db`, `kudiguard stats`, git log

The live sample is **small** (section 5). We state it plainly rather than extrapolate from it.

2. Headline benchmark: frozen set T v1 (`docs/CLAIMS.md` C10, `docs/VALIDATION.md`)

Set. `data/eval/e2e/cases.yaml`: 366 cases (English 147, Pidgin 63, Hausa/Yoruba/Igbo 52 each; 196 scams, 96 genuine messages, 74 questions; 49 scam types), written blind by an independent author and **frozen on 2026-10-10 before any system output was seen**. No rule, prompt, lexicon, retrieval setting or threshold was tuned on it. **Run once**, live, at commit 8e15852, through the real engine with greedy decoding (temperature 0, seed 42) and the product's turn deadline.

Measure	T v1
Pass rate	**85%**
Scams flagged (red or orange)	**178 / 196 (91%)**
Genuine messages wrongly flagged	**35 / 96 (36%)**
Questions answered	72 / 74
Latency (text)	p50 3.5 s, p95 8.3 s
Errors	0
Advice judged blind by 4 LLM judges	good 94, acceptable 108, poor 131, **harmful 33** of 366
Key advice points conveyed (must_advise)	281 / 609 (46%)
Rules-only floor (model off)	88/196 scams flagged, 11/96 genuine flagged

Caveats (as stated in `docs/CLAIMS.md` C10 and `docs/VALIDATION.md`):

- Run once; nothing tuned on T before it.

- **30 of the 33 harmful replies are on genuine messages** (27 called scams): false alarms on real notices are the main failure. About a third of genuine notices are over-flagged.
- Judges are LLMs (one per reply, blind to label/verdict/flags), not people.
- Hausa/Yoruba/Igbo cases are near-parallel templates (not independent samples) and DRAFT wording; the 46 voice cases were run as text.
- **T v2** (re-run at 5503deb, after fixes informed by T v1's failure categories) is **not independent**: same totals (pass 85%, 178/196, 35/96, 72/74; 27 verdicts changed, 6 passes gained and 6 lost), slower (p50 4.3 s, p95 10.9 s); judged good+acceptable 57% -> 60%, harmful 33 -> 36, gains within judge noise; known regression ig-001 (an Igbo vishing SMS now shown green).

Translation analysis on T v1 (`docs/MODEL_STRATEGY.md` §1): the engine flagged 48 of 54 Yoruba and Igbo scams; of the 6 misses, 3 involved translation (2 never translated, 1 cue lost). Translation is not the main cause of missed scams.

3. Advice fixes measured on dev2, before -> after (`docs/CLAIMS.md` C14)

dev2: `data/eval/dev2/cases.yaml`, 130 blind cases (45 genuine, 50 scam, 35 question). Master e9fec66 -> 4abf148 (merged as 5503deb).

Measure	Before	After
Judged good (4 blind LLM judges, mixed)	10	**26**
Good + acceptable	45%	57%
must_advise points conveyed	32%	**46%**
Harmful	25	23
Scams flagged	48/50 (96%)	48/50 (96%)
Genuine flagged	23/45	21/45
Pass	80%	82%
Latency p50 / p95	3.6 s / 8.6 s	4.7 s / 10.2 s

37 cases better, 14 worse. Caveat: **a dev set used to measure the fixes, so not independent evidence**; about 22 of 45 genuine notices are still judged harmful; LLM judges; ha/yo/ig/pcm text DRAFT, n=130.

4. Speech recognition, per language (`docs/SPIKES.md` S6, `docs/CLAIMS.md` C3)

Language	NCAIR1 model	Clips	WER	WER with diacritics	CER
Hausa	NCAIR1/Hausa-ASR	20	28.2%	28.2%	9.0%
Yoruba	NCAIR1/Yoruba-ASR	10	32.7%	64.0%	8.4%
Igbo	NCAIR1/Igbo-ASR	10 (1 rejected, counted as 100% errors)	45.5%	48.9%	25.9%
English	NCAIR1/NigerianAccentedEnglish	1 synthetic clip	0%	—	—

Google FLEURS read sentences re-encoded as WhatsApp audio; not real voice notes; may overlap NCAIR1 training data.

5. Live pilot, 12 Oct 2026

Setup. Live server on the team's PC (RTX 4070 Laptop GPU), N-ATLAS Q4_K_M via llama.cpp, NCAIR1 speech models, public HTTPS via a Cloudflare tunnel, **web voice page** (`/`). Testers recorded voice notes or typed in their

own language about scam situations. WhatsApp was not used by testers on this day (0 WhatsApp users, `kudiguard stats`).

Database totals (`.venv\Scripts\python -m kudiguard.cli stats`, re-run 12 Oct 2026 evening, read-only):

```
Interactions: 27   unique users: 7 (whatsapp users: 0, web sessions: 7)
Languages: {'yo': 15, 'ha': 10, 'en': 1, 'ig': 1}
Verdicts: {'caution': 8, 'no_flags': 8, 'not_a_check': 6, 'likely_scam': 5}
Outcomes: {'answered': 20, 'fallback': 7}
Helpful: 0/0 (n/a)
Latency audio: n=20 p50=7631 ms p95=17686 ms
Latency text: n=7 p50=6136 ms p95=7304 ms
Per day: {'2026-10-08': 1, '2026-10-12': 26}
```

Two of these 27 rows are internal and are excluded below: the team's pre-pilot text test (8 Oct, English) and one team end-to-end smoke test on 12 Oct (Igbo, typed; flagged likely_scam). **Pilot rows, 12 Oct, testers only** (`var/kudiguard.db`, `interactions`, opened read-only; labels from `var/testers.csv`, no names or hashes):

Tester	Language	Voice notes	Typed	Messages
Team lead	Yoruba	4	0	4
Tester 2 (remote)	Hausa	6	0	6
Tester 3 (remote)	Hausa	3	1	4
Tester 4	Yoruba	4	0	4
Tester 5	Yoruba	3	4	7
Total	Hausa 10, Yoruba 15	**20**	**5**	**25**

Verdicts as recorded live (before the same-day fixes): caution 8, no_flags 8, not_a_check 6, likely_scam 3. Outcomes: 18 answered, 7 fixed-text fallbacks. Voice-note latency end to end: n=20, p50 7,631 ms, p95 17,686 ms (`kudiguard stats`; every voice note is a pilot row); speech recognition alone p50 2,364 ms; typed messages median 6,136 ms (n=5). No helpful/unhelpful ratings were given.

How the live system did (team judgement)

There is no official answer key for live messages. The tables below are **the team's judgement of each message** (was the reply right for what the tester meant?).

Tester	Messages	Right	Wrong
Team lead (yo)	4	2	2
Tester 2 (ha)	6	1	5
Tester 3 (ha)	4	3	1
Tester 4 (yo)	4	3	1
Tester 5 (yo)	7	2	5
Total	**25**	**11 (44%)**	**14**

Message type	Count	Outcome
Scam descriptions	16	9 flagged (56%), 7 missed
Off-topic questions	2	2 right (not a check)
Greeting	1	wrong: Yoruba "akúàṣálẹ o" (good evening) translated "I'm in trouble"

Message type	Count	Outcome
Bare numbers	4	shown green: probably the testers' own BVN, prompted by the old text-box example "Someone called and said my BVN is blocked…"; numbers are masked in storage (`[number]`)
Recording glitches	2	wrong: a 0.46 s accidental tap; a speech-model loop ("ìdáná-ìdáná-…" ×60) after which N-ATLAS invented a lottery story

One Tester 4 reply shows the translation risk: from garbled Yoruba, N-ATLAS's English translation invented "the last four digits of my Social Security number" (interaction 19); the rules still gave caution (impersonation of the bank). **Open gap:** a bare number or link gets a generic green answer; KudiGuard cannot verify numbers.

5.1 The real messages (12 Oct 2026, times UTC; transcripts as heard by the NCAIR1 model)

#	Time	Tester	Lang	Input	Transcript (ASR output)	N-ATLAS English	Verdict then
2	16:42	Team lead	yo	voice	ẹnikan pẹrin o so wipe ki mu bifi ẹni mi wa, bo irun bami lati fun ni bifi ẹni mi naa, ṣe skamu lele yii aabi bẹkọ?	— (timed out: model loading)	not_a_check (fallback)
3	16:46	Team lead	yo	voice	ẹnikan te mi, oni ki mu bifi eni mi wa kinsẹndẹ sa wọn, … ki tun fi account-number mi sinu bifi eni mi naa ki wa fẹransẹ si oun	Someone is threatening me …	likely_scam
4	17:00	Team lead	yo	voice	ẹnikọ́pẹ́ mi, o ni ki n fun ohun ni bífi ẹni mi, wọ́n ni bífi ẹni mi ti dina kí n fi ránṣẹ́ sí wọn.	My friend, I have to give something up as a favor …	not_a_check
5	17:01	Team lead	yo	voice	wọ́n ní bífí ẹ̀nì mi ti díná kí n fi ránṣẹ́ sí wọn.	They have blocked my debit card …	caution
6	17:12	Tester 2	ha	voice	shugaban ƙarar ƙarar ƙarar ƙanar	(garbled input)	caution
7	17:13	Tester 2	ha	voice	mun ya kirani ya ce ma tura mai kudi.	they called me and said that they are sending money.	no_flags
8	17:14	Tester 2	ha	voice	wani ya kirani ya ce na tura mai kuɗi.	Someone called me and said they were sending money.	no_flags
9	18:28	Tester 3	ha	voice	wani ya kirani ya ce an toshe bibiyen ɗina.	Someone called and said that my account is blocked.	caution
10	18:29	Tester 3	ha	text	Wani yakirani yace an toshe bvn dina	A stranger is saying that my BVN has been blocked	caution
11	18:50	Tester 2	ha	voice	wani ya ƙira ni ya ce na tura mai otp na.	someone is calling me and said that I sent an OTP	caution
12	18:50	Tester 2	ha	voice	wani ya kirani ya ce na tura mai fim ɗina.	An kira ni don aika fim dinta.	no_flags
13	18:51	Tester 2	ha	voice	wani ya ƙira ne ya ce na tura mai akawun nan ba na.	Someone created a character who said that I sent this accountant here …	not_a_check
14	18:53	Tester 3	ha	voice	wani ya kirani yake ce mun na duba wani sako an turo min wasu lambobi in tura mu shi.	Someone called and said they were checking a message and sent me some numbers to forward to them.	caution
15	18:54	Tester 3	ha	voice	wani ya kira ne yake ce an tura mu wani sako masu wasu lambobi guda shida, to yace in tura masu lambobi.	Someone called and said that there were some numbers with six digits, so he said to send those numbers.	no_flags

#	Time	Tester	Lang	Input	Transcript (ASR output)	N-ATLAS English	Verdict then
16	19:54	Tester 4	yo	voice	bàbá ọ̀rọ̀-ìdáná-ìdáná-ìdáná-… (loop, ~60 repeats)	(invented) "... I have won a big prize of N10 million from the National Lottery"	likely_scam (fallback)
17	19:57	Tester 4	yo	voice	ah, maaamn, roimọt, mo ni kii n ṣọ́fẹ̀ ... kí n ṣẹ̀ndì bbm, ìtì kódì ... there they will get in to my account.	Ah, mom, I'm not lying ...	likely_scam
18	19:59	Tester 4	yo	voice	elórọ̀ imọ́, èyàn kọ́ pé mi ń sín, ó ní kí ń ṣẹndì kọ́dì mi ... wón fẹ́ búlọ̀kà káàntì mi ...	They want me to send them my code first ... they will block my bank account	caution
19	20:08	Tester 4	yo	voice	ẹni kan pe mi ni si ounilati ile ifowopamọ mi loun ti n pe pe wọn fẹ bẹẹmi ...	Someone called me from my bank's customer service ... asked for the last four digits of my Social Security number.	caution
20–23	20:15–20:18	Tester 5	yo	text	[number] (masked in storage; 4 messages)	—	no_flags (fallback)
24	20:19	Tester 5	yo	voice	kí ni àwọn ọmọdé yóò kó yóò bí ní ọjọ́ àwọn ọmọdé?	What will children do on the day of children?	not_a_check (fallback)
25	20:19	Tester 5	yo	voice	akúàṣálẹ̀ o	I'm in trouble	not_a_check
26	20:23	Tester 5	yo	voice	ìgbà wo ni àwọn ọmọ ẹ̀kọ́ tí ó wà?	What time are the students coming?	not_a_check

(# = interaction id. English glosses shortened. "Verdict then" is the verdict recorded live, before the fixes in 5.2.)

What the table shows honestly: the NCAIR1 models heard most Hausa sentences well, but bank words came out phonetically ("bifi ẹni", "bibiyen" for BVN; "fim" for PIN; "akawun" for account), and N-ATLAS translation sometimes flipped the meaning ("send him money" became "they are sending money", interactions 7–8). Several scam situations therefore got no_flags or not_a_check live. These are exactly the cases fixed below.

5.2 Found in live testing -> fixed the same day

	Found	Fix	Commit(s)
a	The first Yoruba voice note timed out (interaction 2: 17.7 s, fallback) because its speech model was loaded on first use	Speech models preloaded at startup (`KUDIGUARD_PRELOAD_LANGUAGES`, `NatlasASR.preload`)	`src/kudiguard/config.py`, `api/services.py`
b	"BVN" heard as "bifi ẹni" (Yoruba) and "bibiyen" (Hausa)	`asr_terms`: per-language regex corrections of phonetically spelled bank terms, applied after ASR; donated audio keeps the recogniser's own words	38e3ac8, 9025acf, 920c415
c	Hausa "ya ce na tura mai kuɗi" (send him money) translated "they are sending money", verdict green	Hausa `money_request` rule on the original text, narrowed to an unknown caller ("wani/wata/wasu ... kira") after review	e736303, 5cca936
d	A 0.46 s accidental tap was transcribed as garbage and answered	Voice notes need at least 0.5 s of voiced speech (`MIN_VOICED_SECONDS`, measured on voiced frames)	e736303, 5cca936
e	Yoruba "give him my BVN" / "my BVN is blocked, send it to them" not flagged	Yoruba caller-framed BVN/NIN/card rules (`asks_credentials`, `account_threat`), only from an unknown caller; bank-branch, NIMC and family requests spared	256b0dc, eaade33, e3f816d

	Found	Fix	Commit(s)
f	Hausa requests for OTP, PIN ("fim"), account number ("akawun nan ba") and a six-digit code not flagged	Hausa rules for those requests from an unknown caller	451629c, 07a1ec6
g	An uncertain non-English voice note with no flags got a green "no warning signs" reply	`KUDIGUARD_VOICE_NO_REASSURANCE` (default true): the reply says some words may have been mis-heard and asks the user to type; stored verdict unchanged	a66a94d, 8d1f551
h	Experiment: a "focused risk check" (an extra N-ATLAS call on the original ha/yo/ig text)	Measured on dev2 (ha/yo/ig/pcm, 70 cases), off -> enforce: scams flagged 25/27 -> 25/27 (no recall gain); genuine flagged 14/25 -> 15/25; Hausa genuine flagged 2/6 (33%) -> 3/6 (50%), n=6; p50/p95 5.5/11.2 s -> 6.5/12.0 s; model calls 187 -> 231. **Rejected** by the pre-set rule (ship only if recall rises and genuine false positives do not): stays off (`KUDIGUARD_FOCUSED_RISK_CHECK=off`), not merged	f961e32 (experiment branch); `data/eval/results/live-2026-10-12/focused-risk-dev2-off.jsonl`, `-enforce.jsonl` (`scripts/eval_e2e.py --summary`)
i	A Yoruba voice note came back as a hyphen-joined loop ("ìdáná-ìdáná-...") and N-ATLAS invented a lottery story from it	Loop guard: hyphen-joined repetitions collapsed; a transcript that is mostly one loop is treated as "couldn't hear clearly"	0038af0
j	Yoruba "ki n sendi kodi" (send my code, English loanwords) not flagged	Yoruba rule added; forwarded "ma fi koodu ranse" (do not send the code) warnings no longer flagged	546c669
k	Testers read the text-box example ("Someone called and said my BVN is blocked...") as a request and typed bare numbers; a Yoruba "good evening" was translated "I'm in trouble"	The text box now asks for the message and says never to type BVN/PIN/OTP; Yoruba evening greetings recognised	17a4e13
l	A real phone number was found in a test file	Replaced by a fictitious number and scrubbed from the published history	b6cfd4e

Tester feedback, acted on. A remote Hausa tester wrote: "*It can't decipher a Hausa pronunciation of words like account. Most Hausa people don't use the Hausa word for account.*" The team followed up: the tester meant the English word "account" in Hausa pronunciation, spelt for example "akawunt". The `asr_terms` corrections now cover Hausa "akawun" / "akawunt" / "akaunt" -> account (and "nan ba" / "namba" after it -> account number) and Yoruba "akaunti" / "akanti" / "akaounti" / "akant" -> account (`data/languages/ha.yaml`, `yo.yaml`; commits 8fc2974, 273c7f6).

How each fix was checked. Each change was reviewed, then false-positive checked on the frozen Hausa and Yoruba cases: **67 Hausa** (52 in T + 15 in dev2) and **67 Yoruba** (52 + 15). Commit messages record the result: Yoruba rules "No change on any Yoruba e2e/dev2 case" (256b0dc, eaade33, e3f816d); Hausa `money_request` "0 new hits on Hausa genuine/question cases" and, after narrowing, "no genuine/question case changes" (one scam case, ha-003, moved from not_a_check to no_flags through the new "someone called me" intent term; 5cca936). All rules remain DRAFT pending native review. The latest full test run, on master 17a4e13/b6cfd4e: **5,533 passed**, 1 skipped, 12 xfailed, **100% line and branch coverage** (9,455 statements, 2,258 branches), mypy clean, ruff clean.

5.3 Replay after the fixes

Today's failed live messages were re-run **as text** through the fixed engine (`scripts/eval_e2e.py`; cases and results in `data/eval/results/live-2026-10-12/`). Verdicts below are read from the result files.

Round 1, Yoruba and Hausa (`replay-yo-ha-cases.yaml` -> `replay-yo-ha-after.jsonl`, 4 cases; the Yoruba texts carry "BVN" as the voice path now writes it after `asr_terms`):

Case (live interaction)	Text replayed	Live verdict	After the fixes
live-ha-2 (7)	mun ya kirani ya ce ma tura mai kudi.	no_flags	**caution** (money_request)
live-ha-3 (8)	wani ya kirani ya ce na tura mai kuɗi.	no_flags	**caution** (money_request)
live-yo-2 (5)	wọ́n ní BVN mi ti díná kí n fi ránṣẹ́ sí wọn.	caution	**likely_scam** (asks_credentials, account_threat)
live-yo-3 (4)	ẹnikọ́pẹ́ mi, o ni ki n fun ohun ni BVN mi, wọ́n ni BVN mi ti dina kí n fi ránṣẹ́ sí wọn.	not_a_check	**likely_scam** (asks_credentials, account_threat)

Round 2, Hausa (`replay-ha-round2-cases.yaml` -> `replay-ha-round2-after.jsonl`, 7 cases): **6 of 7 flagged.**

Case (live interaction)	Text replayed	Live verdict	After the fixes
r2-1 (9)	wani ya kirani ya ce an toshe bibiyen ɗina.	caution	no_flags (see note)
r2-2 (10)	Wani yakirani yace an toshe bvn dina	caution	**caution** (account_threat)
r2-3 (11)	wani ya ƙira ni ya ce na tura mai otp na.	caution	**likely_scam** (asks_credentials)
r2-4 (12)	wani ya kirani ya ce na tura mai fim ɗina.	no_flags	**likely_scam** (asks_credentials)
r2-5 (13)	wani ya ƙira ne ya ce na tura mai akawun nan ba na.	not_a_check	**likely_scam** (asks_credentials)
r2-6 (14)	wani ya kirani yake ce mun na duba wani sako an turo min wasu lambobi in tura mu shi.	caution	**caution** (suspicious_link)
r2-7 (15)	wani ya kira ne yake ce an tura mu wani sako masu wasu lambobi guda shida, to yace in tura masu lambobi.	no_flags	**likely_scam** (asks_credentials)

Note on r2-1: it was replayed as **text**, which skips the voice-only `asr_terms` correction ("bibiyen" -> BVN); its typed twin r2-2 ("an toshe bvn dina") is flagged. As a voice note the correction applies (section 5.2 b).

Read this plainly: the replay uses the same messages the fixes were written from. It shows the fixes work on those messages; it is **not** a measure of general accuracy. General accuracy is T v1 (section 2), and the fixes' false-positive check is the 67 + 67 frozen cases above.

6. Known limitations (stated plainly)

- **Small live sample**: 25 pilot messages on 12 Oct from 5 people (4 external testers and the team lead) on the web page; no WhatsApp users and no ratings yet; the right/wrong split is the team's own judgement. It shows real failure modes, not accuracy.
- **No Igbo or Pidgin real-user messages**: an Igbo tester opened the page but no message arrived (cause unknown, probably on the client side).
- **KudiGuard cannot verify numbers or links**: a bare number or link gets a generic green answer.
- **Hausa ASR ~28% WER** on read speech (Yoruba 33%, Igbo 46%); real voice notes are harder, and bank words come out phonetically. `asr_terms` cover only the terms observed so far.
- **Translation can flip meaning** (live interactions 7–8; 11% of measured translations reversed a negation, `docs/MODEL_STRATEGY.md`). Rules on the original text are the safeguard.
- **About a third of genuine notices are over-flagged** on T (35/96), the main failure; 30 of 33 harmful replies are there.
- **Hausa, Yoruba and Igbo rules, texts and test cases are DRAFT**, pending native-speaker review (none done yet, `docs/VALIDATION.md`).
- Benchmarks use LLM judges, not people; T's ha/yo/ig cases are near-parallel templates.

7. Next (roadmap)

- **Spoken replies in Hausa, Yoruba and Igbo**, requested by testers. Candidates: MMS-TTS (non-commercial licence), YarnGPT (Apache-2.0, 366M parameters), Spitch API.